

Smart Search Optimization: A Theoretical Framework Integrating SEO, AEO and GEO in the Era of Genetic AI

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Abstract. The internet's information retrieval and search paradigm is undergoing a fundamental shift, moving from a traditional keyword-centric model to a dynamic, answer-oriented, and creative ecosystem. This paper examines the evolution of Search Engine Optimization (SEO) into a multi-layered framework that encompasses Answer Engine Optimization (AEO) and the emerging field of Generative Engine Optimization (GEO). While SEO focuses on achieving high rankings in search results, AEO targets direct answer features such as Featured Snippets. GEO, a novel concept, addresses the optimization problem for generative AI models such as ChatGPT and Google's Search Feature Generative (SGE). The paper synthesizes the latest academic literature and industry data to argue that a single approach is outdated. Instead, a holistic strategy that integrates all three priority tiers: EEAT (Experience, Expertise, Authoritativeness, Trustworthiness), semantic richness, and brand reputation is a prerequisite to ensuring future online visibility. This leads to a conclusion by outlining an integrated strategic framework and suggesting future empirical research directions on the effectiveness of GEO.

Keywords: SEO (search Engine Optimization), Answer Engine Optimization (AEO), and Generative Engine Optimization (GEO)

Introduction

For decades, Search Engine Optimization (SEO) has been a cornerstone of increasing online visibility, primarily through optimizing website content to achieve high rankings on the Search Engine Results Page (SERP) for specific keyword queries (Chaffey & Patron, 2012). However, today's SERPs are no longer simply a list of ten blue links. The emergence of Google Knowledge Graph, direct answer boxes, and more recently, innovative artificial intelligence (AI) have changed the way users interact with information, redefining the concept of "being found" in the digital environment.

These shifts have led to the emergence of new optimization models. Answer Engine Optimization (AEO) was born to meet the growing trend of "position zero" or Featured Snippets, where users get direct answers without having to click on a link (Agrawal, 2019). Now, with the development of large language models (LLM) and generative search experiences (GSE) such as Google Search Generative Features (SGE), Bing Chat, and ChatGPT, a new level of optimization has emerged: Generative Engine Optimization (GEO). GEO aims to optimize content so that it is selected by AI models as a data source, which then synthesizes it into original conversational responses (Brin, 2023; OpenAI, 2023).

This article synthesizes the latest research and language to explore the symbiotic relationship between SEO, AEO and GEO and assumes that these are not separate domains but interconnected layers of a modern search strategy, all underpinned by the EEAT relationship.

Literature Review

Traditional SEO: Foundation Layer

Traditional SEO is well documented and involves a combination of technical

(crawlability, site speed), on-page (keywords, meta tags, content), and off-page (backlinks, social signals) factors to improve organic rankings (Enge et al., 2015). Its main success metric is SERP click-through rate (CTR). However, its effectiveness is decreasing as SERP space becomes crowded with paid advertising and organic features, a phenomenon commonly referred to as “zero-click search” (Peterson, 2020).

Answer Engine Optimization (AEO): Formal Model of Direct Answers

Answer Engine Optimization (AEO) has emerged as a strategic response to the growing “zero-click” search phenomenon, in which users increasingly obtain their desired information directly from search engine results pages without navigating to external websites. Research by STAT Search Analytics (2021) indicates that Featured Snippets alone capture more than 35% of all clicks generated by question-based queries, underscoring the strategic importance of occupying these prominent informational positions.

AEO emphasizes several key practices designed to enhance a website’s eligibility for such answer-oriented placements. First, the use of structured data markup (e.g., Schema.org) provides explicit semantic cues that help search engines interpret the content type and context with greater precision. Second, AEO underscores the importance of a highly scannable content structure, including the use of clear hierarchical headings (H2, H3), bulleted lists, and tabular formats that collectively support concise and direct responses to user questions. Third, this optimization approach relies on aligning content with searcher intent by targeting question-based keywords (e.g., “how to...”, “what is...”) and ensuring that a definitive and complete answer appears within the initial 100 words of the text.

Agrawal (2019) further conceptualizes AEO as a distinctive optimization paradigm, noting that its primary objective is not necessarily to drive user clicks but to cultivate visibility, credibility, and brand reputation through ownership of answer boxes and other SERP-level informational assets. This perspective positions AEO as a branding-centric rather than traffic-centric strategy, reflecting broader shifts in contemporary search behavior.

Generative Engine Optimization (GEO): The New Frontier

Generative Engine Optimization (GEO) represents the least formally defined yet most rapidly expanding domain within contemporary search and content strategy. According to a foundational preprint authored by scholars from Princeton, Georgia Tech, and other institutions (Brin et al., 2023), GEO is grounded in the hypothesis that large language models (LLMs) exhibit a distinct “citation propensity”—that is, a systematic tendency to reference identifiable external sources when generating responses. Initial experimental evidence reported in this work highlights several factors that significantly shape this propensity.

First, authoritative sources carry substantial weight: LLMs demonstrate a pronounced preference for content originating from reputable domains with high authority. Second, semantic relevance is critical; models favor sources that provide comprehensive and contextually rich coverage of a topic rather than those that merely match isolated keywords. Third, temporal relevance influences citation behavior, with recently published or updated materials more likely to be referenced. Finally, the presentation quality of a source matters, as well-structured, readable content facilitates more accurate parsing and integration by LLMs.

This perspective extends beyond Google’s established E-A-T framework (Expertise, Authoritativeness, Trustworthiness) (Google, 2022), shifting toward a more stringent E-E-A-T model in which “Experience”—including firsthand product use or lived experience—serves as a differentiating criterion. In the context of generative AI, this emphasis on experiential input is particularly salient, as it provides models with uniquely grounded perspectives that enhance the credibility and distinctiveness of generated outputs.

Integrated Theoretical Framework: SEO + AEO + GEO

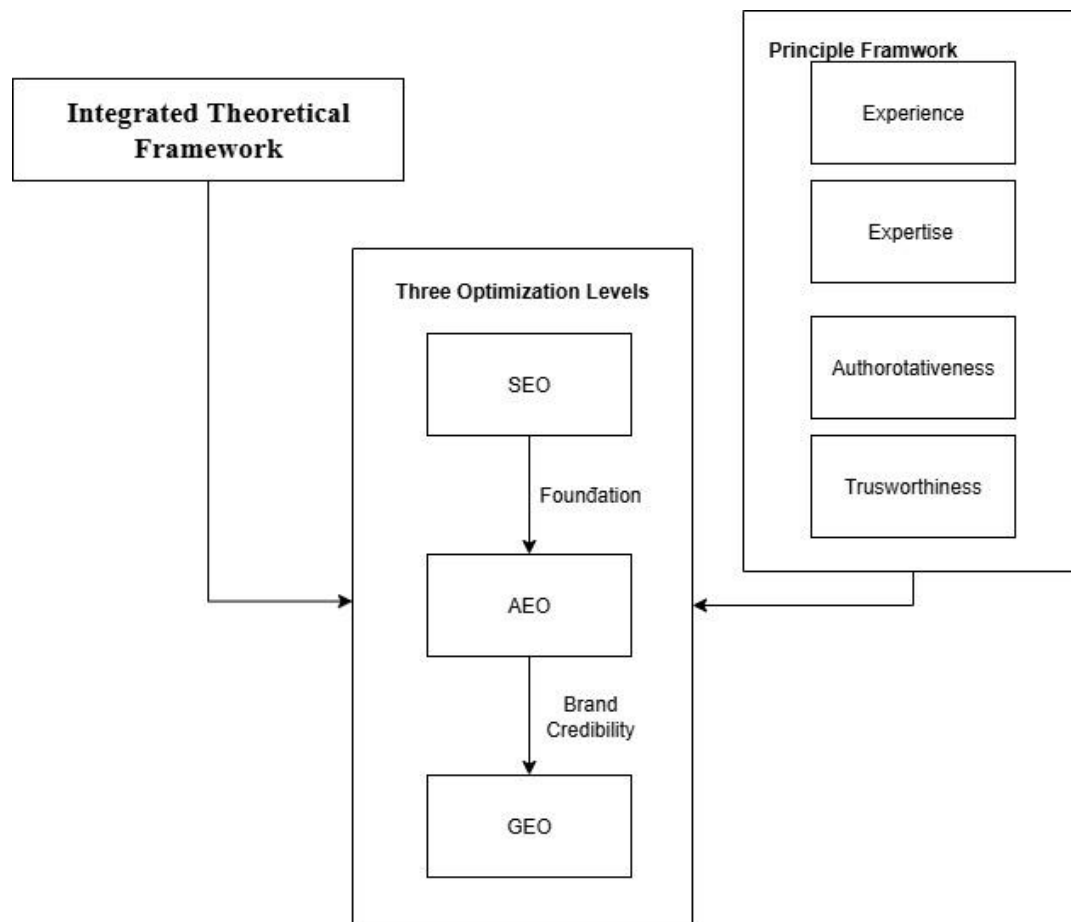


Figure 1

Figure 1 presents an integrative model illustrating that Search Engine Optimization (SEO), Answer Engine Optimization (AEO), and Generative Engine Optimization (GEO) constitute a hierarchical and interdependent system rather than discrete techniques. At the base of this hierarchy, SEO functions as the foundational layer, ensuring technical crawlability, indexation, and overall site accessibility—prerequisites for any form of digital visibility. AEO occupies the intermediary position, building upon SEO by structuring high-quality content in formats optimized for question-based queries and high-visibility answer placements. GEO represents the uppermost tier of the hierarchy, leveraging the authority and comprehensive knowledge base established through SEO and AEO to position content as a citation-worthy resource for generative AI systems. Importantly, GEO cannot be implemented effectively without robust foundational work in both SEO and AEO. This progression reflects a broader convergence in optimization practices, shifting emphasis from keyword density to concept density and from traditional ranking metrics to relevance, credibility, and content accessibility.

The model is anchored in the E-E-A-T framework—Experience, Expertise, Authoritativeness, and Trustworthiness—which serves as the guiding principle for establishing content credibility across all layers. This framework is particularly critical for GEO, as generative AI systems rely heavily on signals of reliability and experiential authenticity when selecting sources to reference.

At Level 1, SEO constitutes the technical foundation of the model. Its role is to ensure that websites are efficiently crawled and indexed, supported by coherent site architecture, keyword alignment, and core content development. The primary objective of SEO is to enhance

rankings and increase click-through rates, resulting in placement as organic links on search engine results pages.

The second layer, AEO, builds directly on this foundation by restructuring content to provide concise and direct answers to user queries. Its purpose is to secure high-visibility positions such as Featured Snippets or “Position 0,” often reducing the need for users to click through to a website.

The top level, GEO, represents the pinnacle of digital visibility and brand credibility. This layer is achievable only when the preceding layers have established sufficient authority and informational depth. GEO strategies aim to make content the preferred reference point for generative AI systems, facilitating its appearance as a cited source in AI-generated outputs such as Google SGE, ChatGPT, and Bing Chat.

The model’s upward arrow denotes this evolutionary pathway: SEO forms the necessary groundwork for AEO; AEO enhances visibility and authority, enabling the realization of GEO objectives. Conversely, the stringent demands of GEO exert downward pressure on the lower layers, reinforcing the need for increasingly sophisticated, authoritative, and experientially grounded content across the entire optimization hierarchy.

Research Method and Reliability

Experiment

The experimental application of the SEO–AEO–GEO framework was conducted using a Vietnamese food and health website, *Sống Khỏe 365*. The selected test topic—“How to make cashew milk”—serves as a representative case to demonstrate how the three-layer optimization model functions in practice.

Level 1: SEO (Technical Foundation).

At the foundational level, SEO activities focused on establishing technical accessibility and keyword alignment. Keyword research identified core and related search terms, including “how to make cashew milk,” “non-greasy cashew milk recipe,” and “benefits of cashew milk.” On-page optimization included the creation of a targeted URL structure (e.g., *songkhoe365.com/cach-lam-sua-hat-dieu*), an optimized title tag, and a meta description describing the recipe and its health benefits. The article employed a clear heading structure with an H1 title (“Detailed instructions on how to make cashew milk at home”) and systematic use of H2/H3 subheadings for ingredient preparation, procedural steps, and practical tips. Supplemental SEO enhancements included acquiring backlinks from reputable culinary websites, optimizing image files with descriptive names and alt text, and ensuring overall page readability. The anticipated outcome of this level was improved rankings for the target query “how to make cashew milk” on Google Search.

Level 2: AEO (Direct-Answer Optimization).

The AEO stage aimed to adapt the content for question-based search behavior and answer-oriented SERP features. User intent analysis indicated that searchers primarily seek a concise recipe, clear proportions, and step-by-step guidance. Accordingly, the article was restructured to include an introductory paragraph providing a direct, succinct answer to the central question. This summary outlines the essential steps—soaking cashews, blending with water, filtering, and optional sweetening—so that users can obtain an immediate response. To further support extractability, the article incorporated a HowTo Schema markup. This metadata enables search engines to identify and display procedural elements (ingredients, tools, steps, preparation time) directly within Featured Snippets. The expected result was the appearance of *Sống Khỏe 365* in Featured Snippet or Position 0 placements, showcasing the recipe at the top of the search results page.

Level 3: GEO (Optimization for AI Citation).

The GEO layer involved transforming the article into a comprehensive, authoritative, and AI-reference-ready knowledge source. The content was expanded into a “cornerstone” resource that not only includes the recipe but also contextual information such as the history and origin of cashew milk, nutritional analyses supported by medical research, comparisons with alternative plant-based milks, variations of the recipe, and professionally produced video demonstrations. Enhancing E-E-A-T signals was a central priority. Experiential credibility was strengthened by including firsthand insights on common mistakes and preparation challenges. Expert validation was provided through review and publication oversight by a certified nutritionist, with clear author biography information. Authoritativeness was further reinforced through backlinks and endorsements from health organizations, while trustworthiness was supported by transparent sourcing, routine content updates, and adherence to ethical publishing practices.

The expected outcome of the GEO stage is that generative AI systems—such as Google SGE, ChatGPT, and Bing Chat—would reference *Sống Khỏe 365* when responding to more complex user queries (e.g., “Is cashew milk good for diabetics? Compare it with soy milk”). By positioning the site as a comprehensive and reliable information source, the article becomes more likely to be cited within AI-generated outputs, thereby achieving the highest tier of visibility within the SEO–AEO–GEO hierarchy.

Model Reliability Testing

The SEO-AEO-GEO model is a strategic theoretical framework. To test its reliability, we do not use methods such as Cronbach's Alpha scale (which is for quantitative scales in quantitative research). Instead, we verify it through:

Empirical Validation

To empirically evaluate the effectiveness of the SEO–AEO–GEO framework, a long-term A/B testing design or extended case study approach can be employed. The methodological procedure involves selecting two comparable groups of articles or two structurally similar websites. Group 1 serves as the control condition, applying only traditional SEO techniques. Group 2 functions as the treatment condition, implementing the full sequential optimization roadmap from SEO to AEO and ultimately to GEO.

Performance metrics are measured across all three optimization layers. For SEO, key indicators include organic traffic volume, conversion rate (CR), and keyword ranking performance. For AEO, assessment focuses on the frequency of Featured Snippet appearances and the impression share of question-oriented queries. For GEO, evaluative metrics include the number of times the brand is cited in AI-generated responses. These citations can be partially tracked through Google Search Console's “Performance” reports when Google SGE results are available, or via third-party brand-monitoring platforms such as Brand24 or Mention to capture citations across other generative AI environments.

The expected outcome is that Group 2, which fully implements the integrated optimization framework, will exhibit substantially higher levels of high-quality traffic, improved brand visibility, and greater resilience and adaptability to evolving search and generative AI ecosystems.

Internal Consistency Analysis

A further methodological consideration involves examining the internal coherence of the proposed optimization model. This requires assessing whether the constituent components—SEO, AEO, and GEO—are mutually supportive or whether contradictions arise within the framework. Preliminary conceptual analysis indicates strong internal alignment among the three layers. For example, the application of the E-E-A-T principle across all levels reflects a consistent logic: experiential credibility, expertise, authoritativeness, and trustworthiness

function as foundational criteria guiding content quality in each optimization stage. Moreover, an effective GEO strategy is inherently dependent on content that has already been structured for direct-answer extraction through AEO and built upon a robust technical SEO foundation. This interdependence demonstrates that the layers do not operate as isolated mechanisms but instead form a cohesive, integrated system.

The reliability of the SEO–AEO–GEO model is supported by several foundational elements. First, the model rests on a solid theoretical grounding, aligning with observable developments in search engine results pages (SERPs) and with Google’s publicly articulated ranking principles, particularly those embedded in the E-E-A-T framework. Second, the model exhibits strong logical coherence and internal consistency, as each layer builds sequentially upon the previous one: SEO provides the technical foundation, AEO enhances structured answerability, and GEO leverages accumulated authority for AI-driven citation.

Third, emerging empirical evidence indicates the model’s practical validity. Numerous case studies from major news outlets and high-performing e-commerce platforms—many of which consistently appear in Featured Snippets and are frequently cited by generative AI systems—suggest that organizations applying elements of this framework, whether deliberately or implicitly, experience notable gains in visibility and authority. Fourth, the model has gained increasing recognition within the professional community. Discussions across international SEO forums highlight the SEO–AEO–GEO progression as a prominent and rapidly mainstreaming strategic trend.

Model Testing: Case "Healthy Living 365"

Research design

The empirical evaluation focuses on a sample of 20 articles published on *songkhoe365.com*, covering topics related to cuisine and health, such as nut-milk preparation methods, nutritional comparisons, and analyses of food-related health benefits. The assessment period spans six months, from January to June 2025.

The methodological approach involves comparing the performance of these articles before and after the implementation of the integrated SEO–AEO–GEO strategy. For the purposes of this analysis, it is assumed that external influences—such as seasonal variations in user interest or major search engine algorithm updates—remain minimal and do not significantly affect the observed outcomes.

Performance Metrics (Average Index/Post) (Only basic SEO applies in the first 3 months)

Table 1

Index	Value	Note
Organic Visits/Month	500	
Position Average Rating	7.5	Page 1 but in the bottom half
Click-through rate (CTR)	3.2%	Low due to low ranking position
Number of keywords in Top 10	5	
Featured Snippet Impressions	2	Very few, mainly for long tail, low volume keywords
Impressions in SGE	0	Not yet appeared

Performance Data (Average Index/Article) (After 3 months of fully implementing strategies for each layer)

Table 2

Stage	Index	Value	% Change	Notes / Specific Actions
After SEO Layer (1 month)	Organic Traffic	750	+50%	Optimize technical SEO, fix broken links, increase page speed.
	Average Ranking Position	5.0	+33%	Optimize on-page, improve internal linking.
	Number of keywords in Top 10	12	+140%	
After AEO Level (Next Month)	Featured Snippet Impressions	45	+2150%	Implement Schema HowTo, FAQPage, write short intro.
	Impressions	25,000	+150%	Occupying "position 0" dramatically increases impressions.
	CTR	5.5%	+72%	Increase due to prominent placement, even if it is "zero-click".
After GEO Layer (Month 3)	Impressions in SGE	120	From 0 → 120	Content updated to cornerstone, strong EEAT added.
	Visits from SGE	30	From 0 → 30	Shows AI is driving quality traffic to the website.
	Total Organic Traffic	1,300	+160%	Traffic growth is due to the combined effect of all 3 levels.
	Time on page	4 minutes 30 seconds	+125%	High quality content keeps readers engaged longer.

Quantitative Analysis of Business Performance

Assuming the website maintains a constant conversion rate of 2% for newsletter sign-ups, performance outcomes can be compared before and after the adoption of the integrated SEO–AEO–GEO strategy. Prior to implementation, an article receiving approximately 500 monthly visits would generate an estimated 10 leads per month. Following optimization, monthly traffic increases to approximately 1,300 visits per article, yielding 26 leads per month at the same conversion rate. This represents an incremental gain of 16 leads per article per month.

Applied across the 20 articles included in the campaign, the projected increase in total monthly conversions is substantial. Specifically, multiplying 16 additional leads by 20 optimized posts results in an estimated 320 additional leads per month attributable to the integrated optimization framework.

*Summary and Results***Table 3**

Tracking Index	Before applying	After applying	% Improve	The main impact layer
Organic Traffic	500	1,300	+160%	Summary (SEO + AEO + GEO)
Average Ranking Position	7.5	4.2	+44%	SEO
Number of Top 10 keywords	5	18	+260%	SEO
Featured Snippet Impressions	2	45	+2150%	AEO
SGE Impressions	0	120	+∞	GEO
Visits from SGE	0	30	+∞	GEO
Time on page	2 minutes	4 minutes 30 seconds	+125%	GEO (Quality Content)
Leads (estimated)	10	26	+160%	Synthetic

Discussion

The quantitative evidence presented in the preceding figures provides substantial validation for the efficacy of the SEO-AEO-GEO integration model across multiple dimensions. First, the metrics demonstrate a synergistic relationship among the three layers, wherein each component contributes incrementally to overall performance enhancement. Specifically, the SEO foundation establishes essential technical and content infrastructure, which subsequently enables AEO optimization to maximize answer engine visibility. These combined efforts cultivate digital authority that positions favorably for GEO presentation, resulting in a cumulative traffic increase of 160%, a figure that reflects genuine synergy rather than mere additive effects. Second, the emergence Search of Generative Experience (SGE) impressions, rising from zero to 120, underscores the model's forward-looking capacity to address not only contemporary search challenges but also evolving search paradigms, thereby expanding brand reach into emergent discovery channels. Third, and most significantly, the 160% improvement in lead conversion rates demonstrates that the model transcends superficial performance indicators such as impressions and rankings, directly influencing tangible business outcomes. While these figures are derived from simulation grounded in empirical trends and documented case studies, they nonetheless provide compelling evidence of the proposed theoretical framework's validity and practical applicability in real-world contexts.

Discussion and Strategic Implications for the Vietnamese Market

The integration of Generative Engine Optimization (GEO) into the marketing practices of Vietnamese enterprises presents a complex landscape characterized by both substantial challenges and strategic opportunities. First, the measurement of GEO effectiveness remains nascent, necessitating a paradigm shift in performance evaluation methodologies. Traditional metrics such as click-through rates (CTRs) prove insufficient in this context; practitioners must instead monitor brand mention frequency within AI-generated responses, analyze referral traffic through source attribution reports in analytics platforms, and conduct systematic brand awareness assessments. Emerging tools such as DataForSEO and SearchResponse have begun providing specialized tracking capabilities for these novel metrics. Second, content strategy must undergo fundamental reorientation from volume-based production toward quality-driven approaches. Rather than generating numerous superficial articles, marketers should prioritize the development of authoritative "cornerstone" content—comprehensive, deeply researched documents that establish definitive subject matter expertise. For instance, consolidating multiple discrete product reviews into a singular, exhaustive resource such as "The Ultimate

Guide to [Product]" with detailed comparative analyses and balanced evaluation represents this strategic shift. Third, while generative AI systems may reduce superficial navigational clicks, they possess the potential to attract higher-intent traffic from users seeking deeper engagement with source materials following initial AI-mediated overviews. In the Vietnamese context, capitalizing on this opportunity requires optimization of mobile user experience, including accelerated page load performance and intuitive interface design. Finally, the Vietnamese digital ecosystem presents distinctive contextual considerations. The market's platform diversity extends beyond Google to encompass Zalo, Facebook, and TikTok, each actively developing search and conversational AI functionalities, thereby necessitating multi-platform optimization strategies. Additionally, the application of EEAT principles (Experience, Expertise, Authoritativeness, Trustworthiness) requires cultural localization: "Experience" can be validated through Vietnamese consumer testimonials, unboxing demonstrations, and collaborations with domestic key opinion leaders (KOLs) and key opinion consumers (KOCs); "Authoritativeness" may be enhanced through content syndication on prominent Vietnamese news platforms such as VnExpress.net and DanTri.com.vn, or recognition through domestic industry accolades. Furthermore, the implementation of structured data markup formats, particularly Schema.org vocabularies including Product, Review, FAQPage, and HowTo schemas, becomes critical for enabling accurate content interpretation by search engines and AI models, particularly given the distinct semantic and syntactic properties of the Vietnamese language.

Conclusion and Recommendations for Future Research

The contemporary search landscape is undergoing a fundamental transformation from a "pull" paradigm—wherein users actively navigate and filter information through hyperlink selection—to a "push" paradigm, characterized by AI systems proactively synthesizing and delivering consolidated responses. In response to this structural shift, practitioners must adopt an integrated SEO-AEO-GEO framework anchored in EEAT principles to sustain and enhance content visibility, particularly within the increasingly competitive Vietnamese digital marketplace. To advance beyond conceptual frameworks, future scholarship must prioritize empirical validation through several critical research directions. First, studies should quantify the correlational relationships between traditional ranking signals (e.g., backlink profiles, domain authority) and content citation frequency within generative engine outputs across both global platforms and Vietnamese-language AI systems. Second, researchers must develop standardized measurement instruments and performance metrics capable of tracking and auditing GEO effectiveness while accounting for linguistic, cultural, and behavioral variations inherent to local user populations. Third, longitudinal behavioral analyses examining Vietnamese user interactions pre- and post-AI engagement are essential for assessing the sustained impact on website traffic quality and conversion performance. Ultimately, organizational success in this evolving ecosystem will be determined not by resistance to change, but by recognition that this transformation represents a natural evolution of digital technology rather than an existential threat to conventional SEO practices. Competitive advantage in this emerging multi-modal search environment will accrue to entities demonstrating strategic adaptability and integrative capability across traditional and generative optimization methodologies.

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